

JOONWON LEE

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Google, YouTube Marketing
San Bruno, CA

Re: Business Data Scientist, YouTube Marketing

Dear YouTube Marketing Hiring Team,

I am applying for the Business Data Scientist position with YouTube Marketing. I am a Statistics Ph.D. candidate at Rutgers University, expecting to graduate in December 2026, and hold an M.A. in Economics. My background combines hypothesis testing, regression analysis, causal inference, and machine learning - methods central to measuring marketing impact and translating evidence into business recommendations.

At JPMorgan Chase, I developed and validated a Sequential Probability Ratio Test for binary risk indicators, converting a monitoring question into testable hypotheses and measurable decision rules. I designed simulation studies to evaluate Type I and Type II error, detection delay, and early-decision tradeoffs, then communicated the model's behavior and limitations. At Travelers, I developed an end-to-end LightGBM gradient-boosting pipeline across more than 2.46 million property-risk records and benchmarked it against generalized linear regression. I evaluated model performance and feature attribution and presented the findings as pricing and risk-segmentation recommendations.

My economics research provides direct causal-inference experience. I estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household covariates. This work strengthened my ability to define treatment and outcome measures, construct an appropriate comparison group, address confounding, and interpret estimated effects within the assumptions of an observational study.

These experiences prepare me to design rigorous measurement studies, use regression and machine learning to identify drivers of impact, and communicate clear recommendations to Product and Marketing partners. I would welcome the opportunity to apply this combination of statistical rigor and practical Python modeling to YouTube's marketing and user-growth questions. Thank you for your consideration.

Sincerely,
Joonwon Lee